

peak value of inrush current is given on y-axis. Peak value of inrush current for each cycle is written on the graph itself. Brief analysis of the transformer's inrush current characteristics also pops up on the screen.

Inrush current is calculated on per-phase basis. In case of a 3-phase transformer, during the inrush condition, only one phase has large inrush current—that is, the phase that gets switched at the worst or near worst instant of voltage switching (meaning when switching angle is 0 degree or near 0 degree). The software calculates this value of inrush current.

Calculate-Effect of Switching Angle. If this option is selected, value of the first peak of inrush current is displayed in a table with respect to switching angle of the source voltage for the worst case, that is, when residual flux is in the direction of flux change. Inrush current value is displayed for switching angle of 0 degree, 30 degrees, 60 degrees and so on up to 180 degrees in a difference of 30 degrees. It can be observed from these values that inrush current decreases when switching angle increases.

Harmonic analysis. If this option is selected, the following information is displayed on the screen in tabular form:

1. Peak value of inrush current for DC component (0Hz), fundamental frequency, harmonics of second order up to tenth order and individual distortion ID_n (magnitude of the nth harmonic as a percentage of the fundamental)
2. Total harmonic distortion (THD) in percentage

Software testing

Software has been tested for various examples available in standard textbooks of transformer engineering

including the test problem example given below:

Transformer data.

1. Capacity: 31500kVA
2. Vector group: 3-phase, Star/Delta
3. Primary/excited winding voltage: 132kV (value entered on per-phase basis: $132/\sqrt{3}$, i.e., 76.21kV)
4. Secondary winding voltage: 33kV
5. Power frequency: 50Hz
6. Number of turns of primary/excited winding: 920
7. Primary/excited winding height: 1250mm

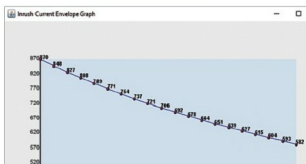


Fig. 6: Inrush current graph

Sl. No.	Switching Angle(Degree)	Ip(peak)(Amp)
1	0deg	858.36
2	30deg	786.75
3	60deg	591.1
4	90deg	323.85
5	120deg	56.6
6	150deg	-139.05
7	180deg	-210.66

Fig. 7: Switching angle

Sl. No.	Harmonic Order	Ip(peak)(Amp)	% Harmonics(%)
1	DC	265.45	-
2	Fundamental	461.37	-
3	2	57.21	12.4
4	3	34.73	7.53
5	4	13.8	2.99
6	5	7.3	1.58
7	6	3.28	0.71
8	7	1.48	0.32
9	8	0.78	0.17
10	9	0.43	0.09
11	10	0.25	0.05

Fig. 8: Harmonic analysis

8. Primary/excited mean dia: 980mm

9. Total resistance (including system resistance): 0.9 ohm

10. Peak flux density (Bm): 1.7 Tesla

11. Saturation flux density (Bs): 2.03 Tesla

12. Core material: Cold rolled steel

13. Core area: 220,000 sqmm

14. Number of cycles for plotting inrush current graph: 20

Note. For cold rolled material, residual flux density is 80 per cent of the peak flux density, whereas for hot rolled material, it is 60 per cent of the peak flux density. Therefore in this case $B_r = 0.8B_m$, that is, 1.36 Tesla

Results

Inrush current graph. Here the inrush current graph is plotted. The number of cycles is given on x-axis, while peak value of inrush current is given on y-axis. Peak value of inrush current for each cycle is written on the graph itself. The graph is plotted for the selected number of cycles, that is, 20. This window is shown in Fig. 6.

Calculate-Effect of Switching Angle. Results are displayed in tabular form as shown in Fig. 7.

It can be seen from these results that as the switching angle increases, the value of peak decreases.

Harmonic Analysis. Results are displayed in tabular form as shown in Fig. 8.

Total Harmonic Distortion (THD): 15.14%

It is obvious from above results that the second harmonic is the most dominating harmonic component in inrush current and accounts for almost 80 per cent of THD.

Effect of various transformer parameters on inrush current

This software can give a useful insight